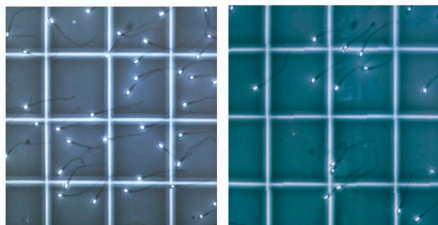


PROBLEMS WITH SPERM PRODUCTION

A man may produce sperm in low quantities, or his sperm may be deformed or unable to swim properly. All these problems reduce the likelihood that his sperm can contribute to conception. Huge numbers of sperm must be produced in order for fertilization to occur; men in whom this does not happen have a low sperm count. Microscopic examination can reveal this problem and can also look at the size, shape, and movement (motility) of individual sperm. Problems in any of these areas can cause reduced fertility. If only a small volume of semen is produced per ejaculation, fertility may also be reduced.



NORMAL SPERM COUNT

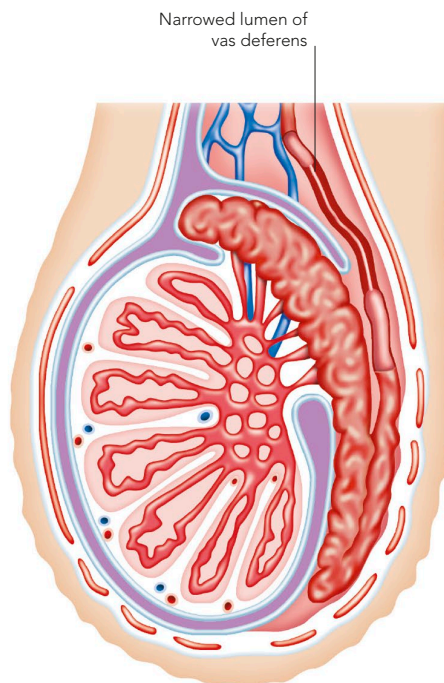
LOW SPERM COUNT

EJACULATION PROBLEMS

A number of ejaculation problems prevent sperm from arriving in the vagina by the normal means, making fertilization impossible. The most common is erectile dysfunction (the difficulty in achieving or maintaining an erection). This condition may be a result of diabetes mellitus (see pp. 142–43), a spinal cord disease, impaired blood flow, certain drugs, or psychological problems. Another problem, retrograde ejaculation, causes semen to flow back into the bladder because of faulty valves; this can be a complication of surgery for partial or complete removal of the prostate gland. Various treatments are available that can help to reduce erectile dysfunction, depending on the nature and cause of the problem.

DIFFICULT PASSAGE OF SPERM

Sperm has a long and tortuous journey from its source in the testis until it is ejaculated. Narrowing, blockage, or other distortion of any of the tubes, including the epididymis and vas deferens, that make up this network can slow or completely block the passage of sperm. Causes of this problem are various, but infection of the male reproductive system is most likely. Some sexually transmitted infections (STIs, see p.298), most notably gonorrhoea, can cause inflammation of the tubes, which leaves scar tissue that can distort their structure and affect their sperm-carrying ability.



INFLAMED VAS DEFERENS

Damage to the vas deferens, one of the tubes that transports the sperm, can prevent or slow down its passage. Infection, usually by a sexually transmitted organism, can be responsible for such damage.